

AKADEMIKA

ACL-FS

**FOURIER SYNTHESIS TRAINER
SYSTEM**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. ANALOG COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

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SAFETY RULES

Carefully follow the instructions contained in this manual as they provide you with important points on safety during the installation, use and maintenance. Keep this manual always with you for easy reference.

Arrange all accessories in order after unpacking, so that their integrity is checked with respect to the packing list. Also ensure that no visible damage as such appear on any accessories.

Before connecting the power supply to the equipment, be sure that the jumpers and connecting chords are connected correctly as per experiment.

The power supply cable must be set so that they cannot be trodden upon or squeezed by objects.

On the equipment, there are some slots or opening for the ventilation: to ensure a reliable operation and to protect the equipment from overheating, they must not be blocked or covered. This equipment must be in such a position to enable a proper aeration.

Do never set the on trolleys, supports, tripods, stirrups or unstable tables. The equipment could fall causing damage to the collided person or it can damage itself. Any installation of the equipment must follow the instructions of the manufacturer and must be carried out using recommended accessories.

This equipment must be employed only for the use it has been conceived, i.e. as educational equipment, and must be used under the direct survey of expert personnel. Any other use is not proper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

Precautions:-

In order to safeguard the user's safety and the equipment operation, when using electrical equipment some fundamental rules must be followed. In particular the following regulation for use must be followed:-

Ambient temperature: from 0 to 45°C

Relative Humidity: from 20 to 80%

Avoid any quick shift of temperature and humidity.

In case of fault and or bad operation, turn off the equipment and do not tamper it. In case of reparation, ask the center for technical assistance or ask exclusively original spare parts. If these conditions are not respected, the equipment can be compromised.

In case of penetration of objects or liquids inside the equipment, disconnect the power supply cable and make it checked by qualified personnel before using again.

Cleaning the Equipment

Use a dry and soft cloth to clean the container and the silk screen panel. Do never use insecticide or chemical products or solvents for cleaning.

Vibration or Collisions

Be careful not to cause vibrations or collision

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well. For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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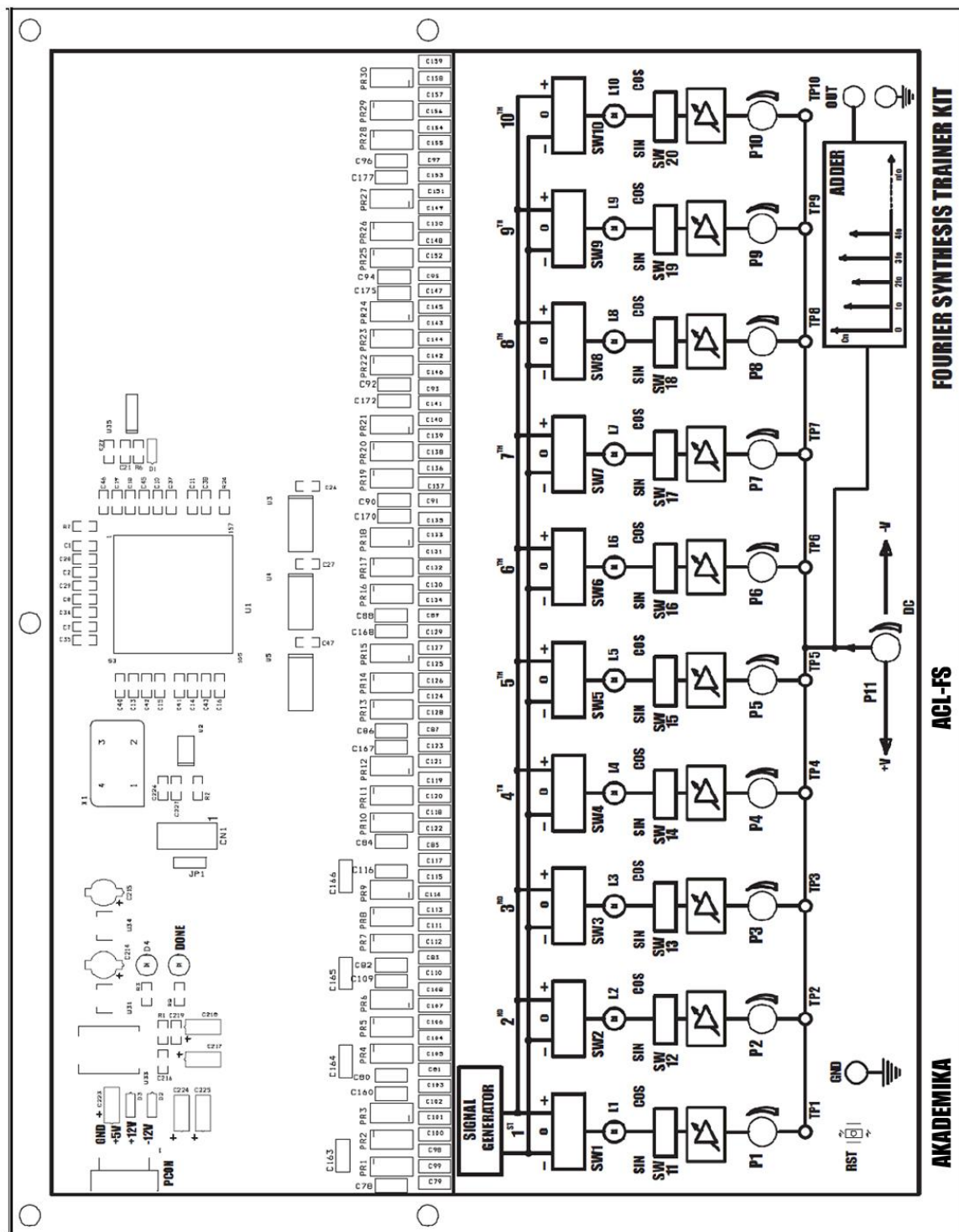
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TECHNICAL SPECIFICATIONS

- 10 sinusoidal generators (one dominant wave at 10 KHz + 9 harmonic waves 20/30/40/50/60/70/80/90/100 KHz)
- 10 three-position two-way switches (one each sine curve) for the selection of: direct signal (+), reverse (-), null (0)
- 10 two-way switches (one each signal) for the selection either of the sine function or of the cosine one.
- 10 amplifiers (one each signal) having a release –adjustable gain (1 to 10 Vpp max.)
- A direct voltage generator from -2.5V to +2.5V
- An adder stage that sums up the ten sine signals and the direct voltage.
- Adjustable amplitude from 0 to 10 Vpp
- Accessories

ACL-FS			Quantity
1	Short Links PCS2 Patch cord	Male To Male	2
2	ACL-FS EXPT Manual		1
3	Power Supply		1
4	Power cord		1

FUNCTIONAL BLOCK



INTRODUCTION

Fourier synthesis is a method of electronically constructing a signal with a specific, desired periodic waveform. It works by combining a sine wave signal and sine-wave or cosine-wave harmonics (signals at multiples of the lowest or fundamental, frequency) in certain proportions. The scheme gets its name from a French mathematician and physicist named Jean Baptiste Joseph, Baron de Fourier.

Many waveforms represent signal energy at a fundamental frequency and also at harmonic frequencies (whole-number multiples of the fundamental). The relative proportions of energy concentrated at the fundamental and harmonic frequencies determine the shape of the wave. The wave function (usually amplitude, frequency, or phase versus time) can be expressed as of a sum of sine and cosine functions called a Fourier series, uniquely defined by constants known as Fourier coefficients. If these coefficients are represented by $a_0, a_1, a_2, a_3, \dots, a_n, \dots$ and $b_1, b_2, b_3, \dots, b_n, \dots$, then the Fourier series $F(x)$, where x is an independent variable (usually time), has the following form:

$$F(x) = a_0/2 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \dots \\ + a_n \cos nx + b_n \sin nx + \dots$$

In Fourier synthesis, it is necessary to know, or to determine, the coefficients $a_0, a_1, a_2, a_3, \dots, a_n, \dots$ and $b_1, b_2, b_3, \dots, b_n, \dots$ that will produce the waveform desired when "plugged into" the generalized formula for the Fourier series, as defined above. Then, sine and cosine waves with the proper amplitudes (as defined by the coefficients) must be electronically generated and combined, up to the highest possible value of n . The larger the value of n for which sine-wave and cosine-wave signals are generated, the more nearly the synthesized waveform matches the desired waveform.

Fourier synthesis is used in electronic music applications to generate waveforms that mimic the sounds of familiar musical instruments. It is also employed in laboratory instruments known as waveform generators or function generators. These devices are used to test communication systems.

Any periodic function, no matter how trivial or complex, can be expressed in terms of converging series of combinations of sine and/or cosines, known as Fourier series. Therefore, any periodic signal is a sum of discrete sinusoidal components.

The Fourier theorem is fairly general and also applies to periodic functions that have discontinuities and cannot be represented by a single analytic expression. For a periodic function $f(x)$, provided that the following conditions are satisfied (Dirichlet conditions):

- (a) $f(x)$ is defined and single-valued except (perhaps) at a finite number in $(-T, T)$,
- (b) $f(x)$ is periodic outside $(-T, T)$ with period $2T$
- (c) $f(x)$ and $f'(x)$ are piecewise continuous in $(-T, T)$, then $f(x)$ can be expressed by the following series:

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{T} + b_n \sin \frac{n\pi x}{T} \right)$$

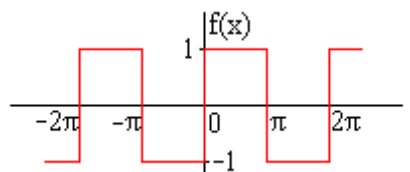
Where

$$a_n = \frac{1}{T} \int_{-T}^T f(x) \cos \frac{n\pi x}{T} dx$$

$$b_n = \frac{1}{T} \int_{-T}^T f(x) \sin \frac{n\pi x}{T} dx$$

Although the calculation of a_0 , a_1 , b_1 , a_2 , b_2 is a mathematically straightforward process, it may become rather tedious depending on the complexity and the discontinuities of $f(x)$. The Fourier theorem is particularly useful in scientific instrumentation, where $f(x)$ functions may represent actual periodic signals $f(t)$. Some simple examples of Fourier series are those of square, triangular and saw-tooth waveforms:

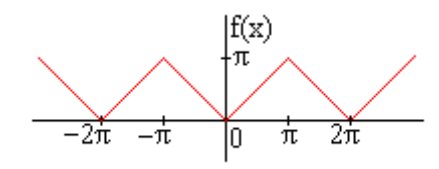
Square waveform



$$f(x) = |x| = \begin{cases} 1 & 0 < x < \pi \\ -1 & -\pi < x < 0 \end{cases} =$$

$$= \frac{4}{\pi} \left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \dots \right)$$

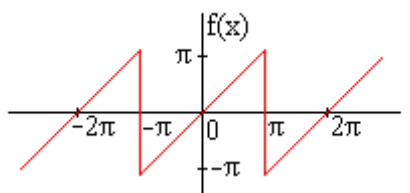
Triangular waveform



$$f(x) = |x| = \begin{cases} x & 0 < x < \pi \\ -x & -\pi < x < 0 \end{cases} =$$

$$= \frac{\pi}{2} - \frac{4}{\pi} \left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right)$$

Saw-tooth waveform



$$f(x) = x, \quad (-\pi < x < \pi) =$$

$$= 2 \left(\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \dots \right)$$

Fourier Synthesis

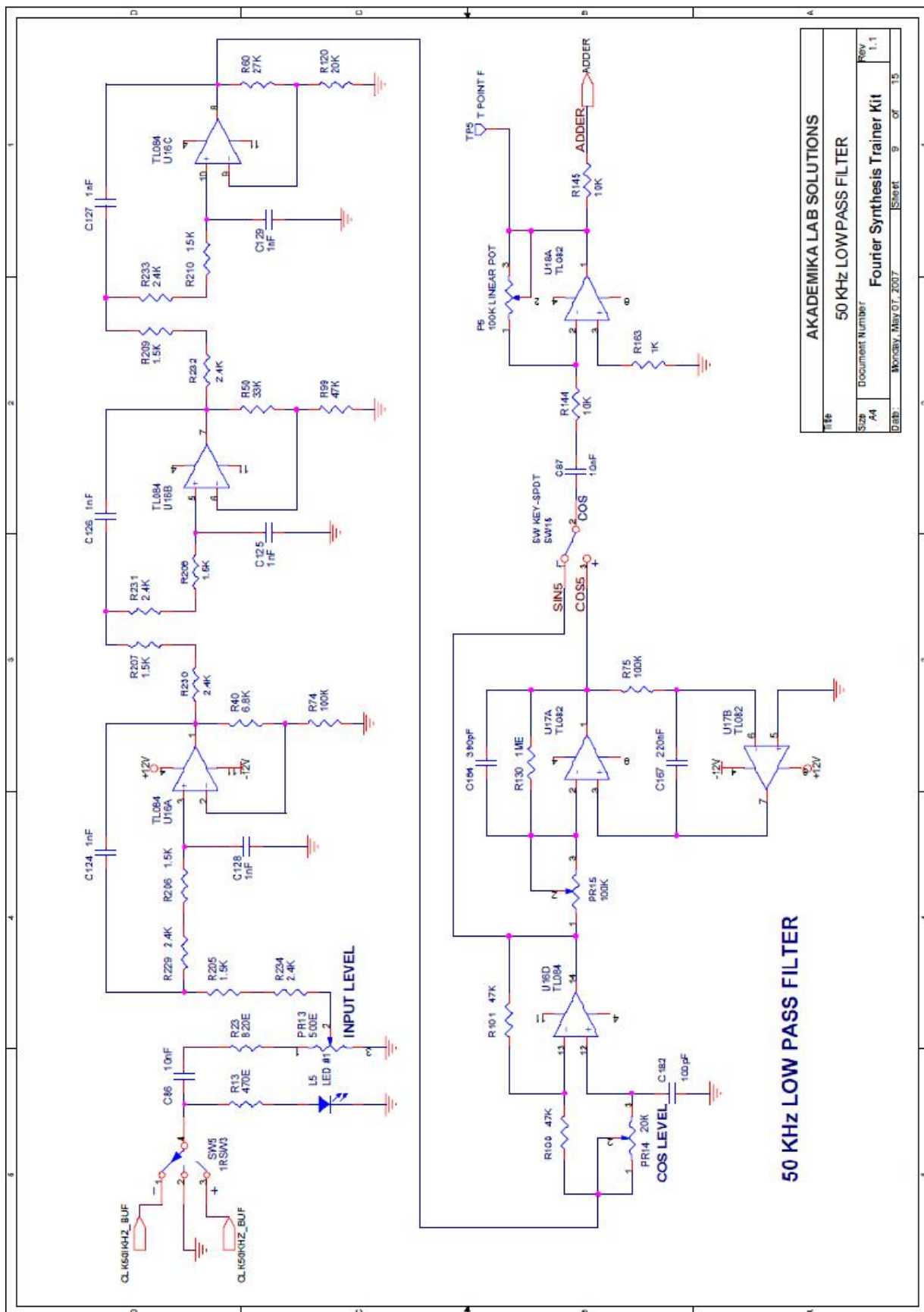
Fourier synthesis is based on sine waves & its harmonics. Thus to generate the sine wave & its harmonics the power of FPGA technology is used. IC U31 (FPGA-XC3S400) & U2 (PROM -XCF04S) is programmed to generate 10 square waves being synchronous among them, whose frequencies are 10 KHz, 20 KHz, 30 KHz, 40 KHz, 50 KHz, 60 KHz, 70 KHz, 80 KHz, 90 KHz and 100 KHz.

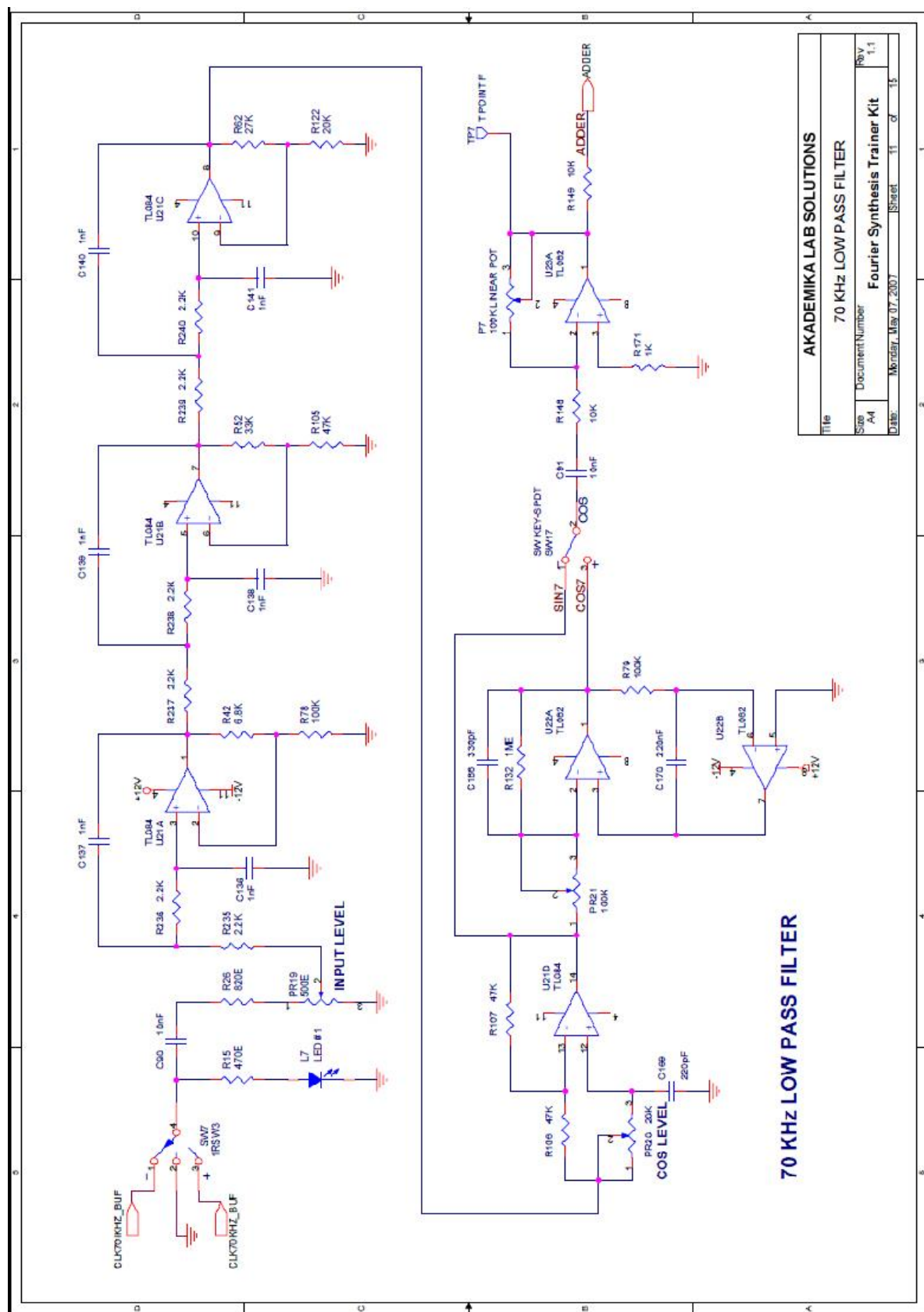
Each square wave is then filtered so as to obtain the sine wave. Ten different fourth order Butterworth filters are designed for each frequency. All pass filters are also designed to get the 90° phase shift so as to get the cosine waveforms.

Finally the adder is built around CA3140 U32 to add the selected harmonics.

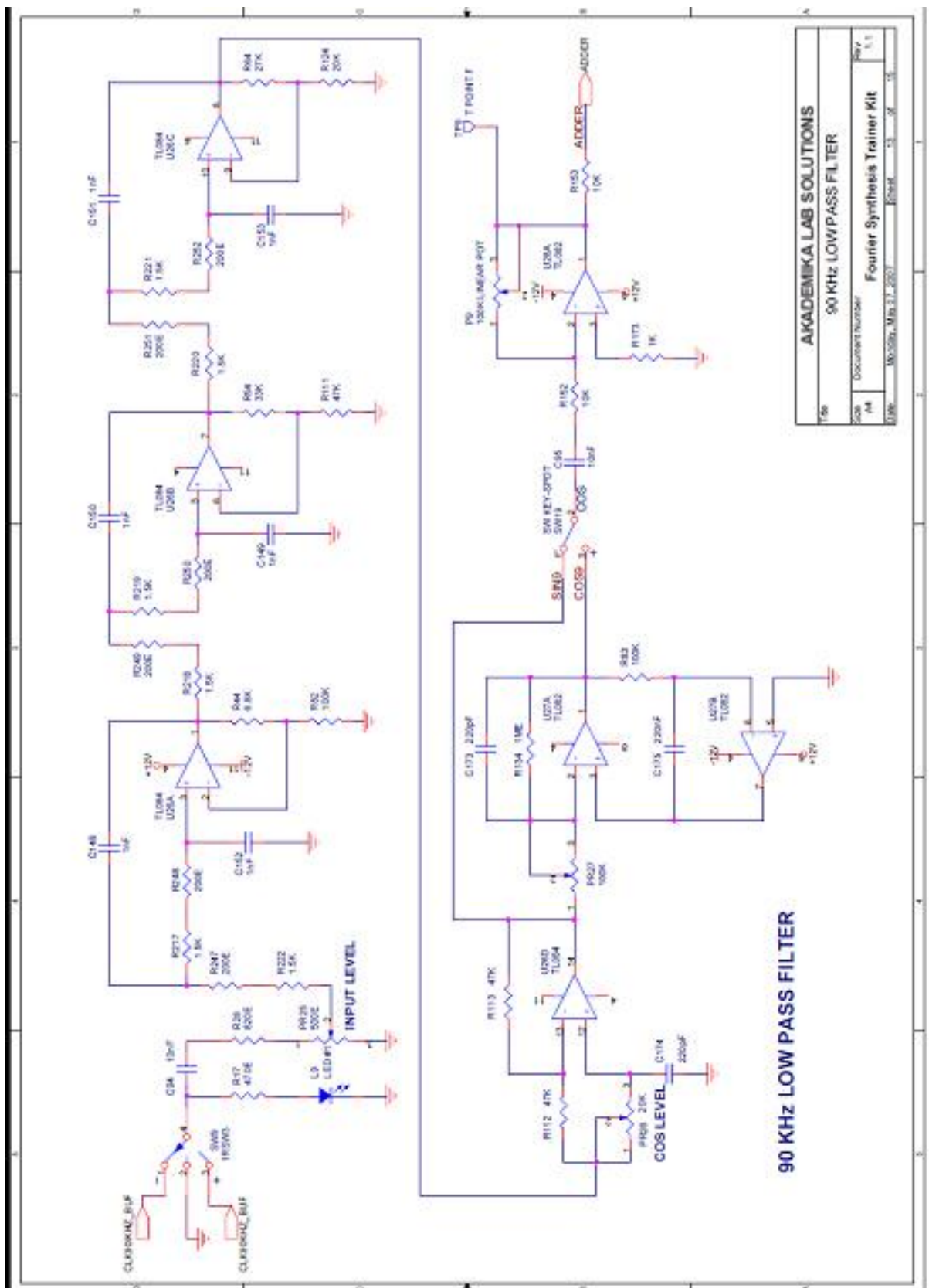
To adjust the amplitude of each sine wave 10 different amplifiers are used, the gain of each amplifier can be varied from 1Vp-p to 10Vp-p.

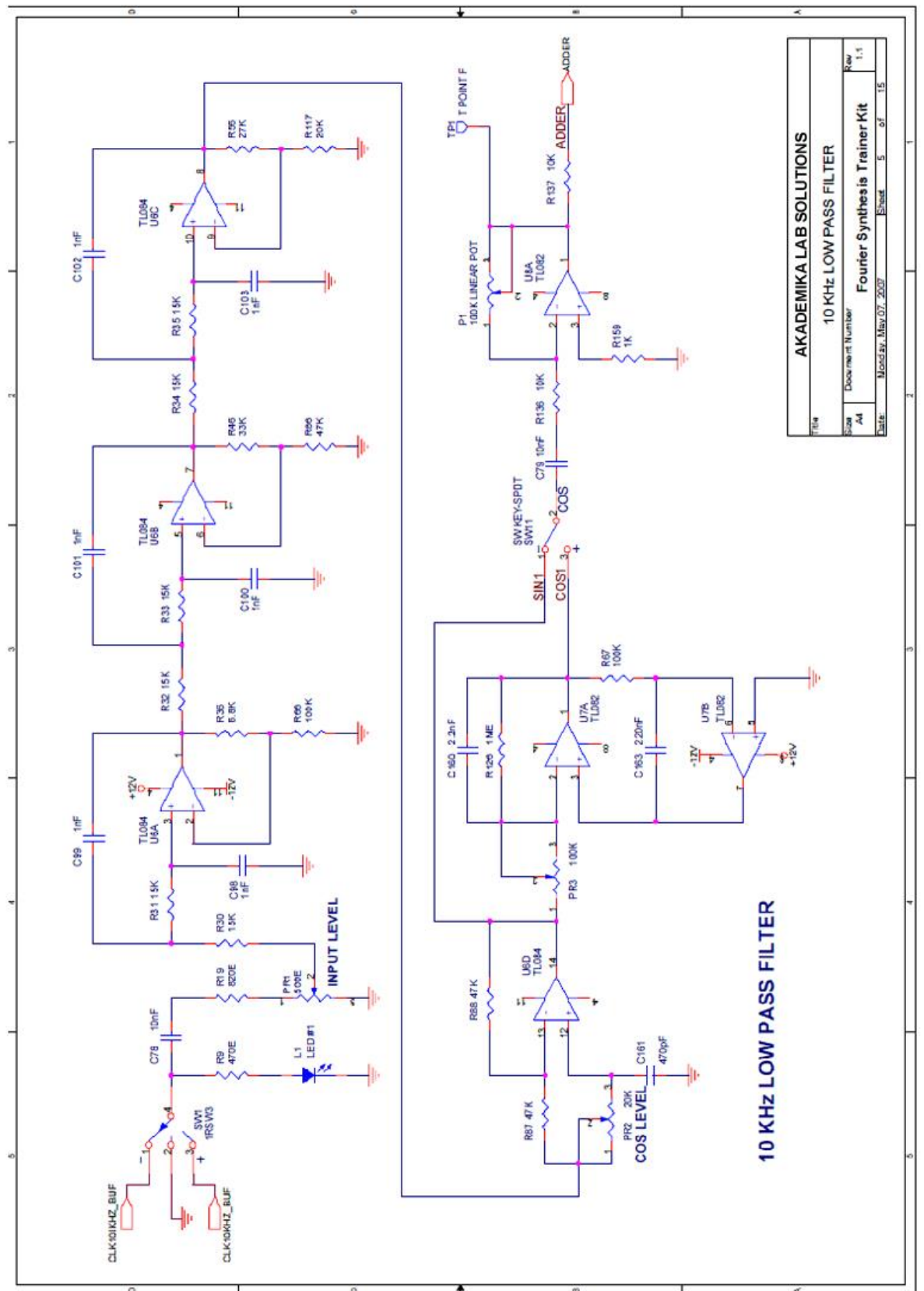
CIRCUIT DIAGRAM

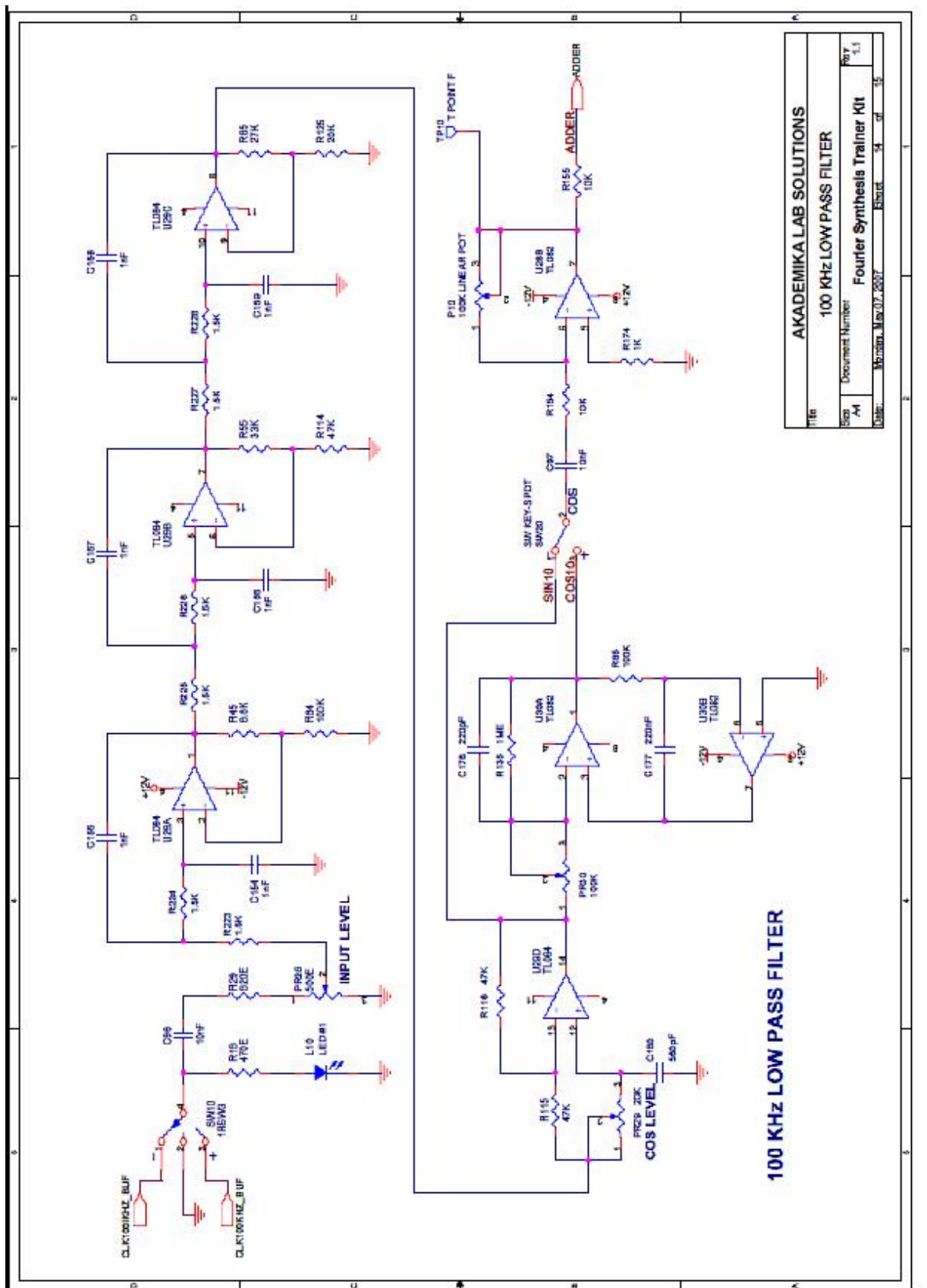


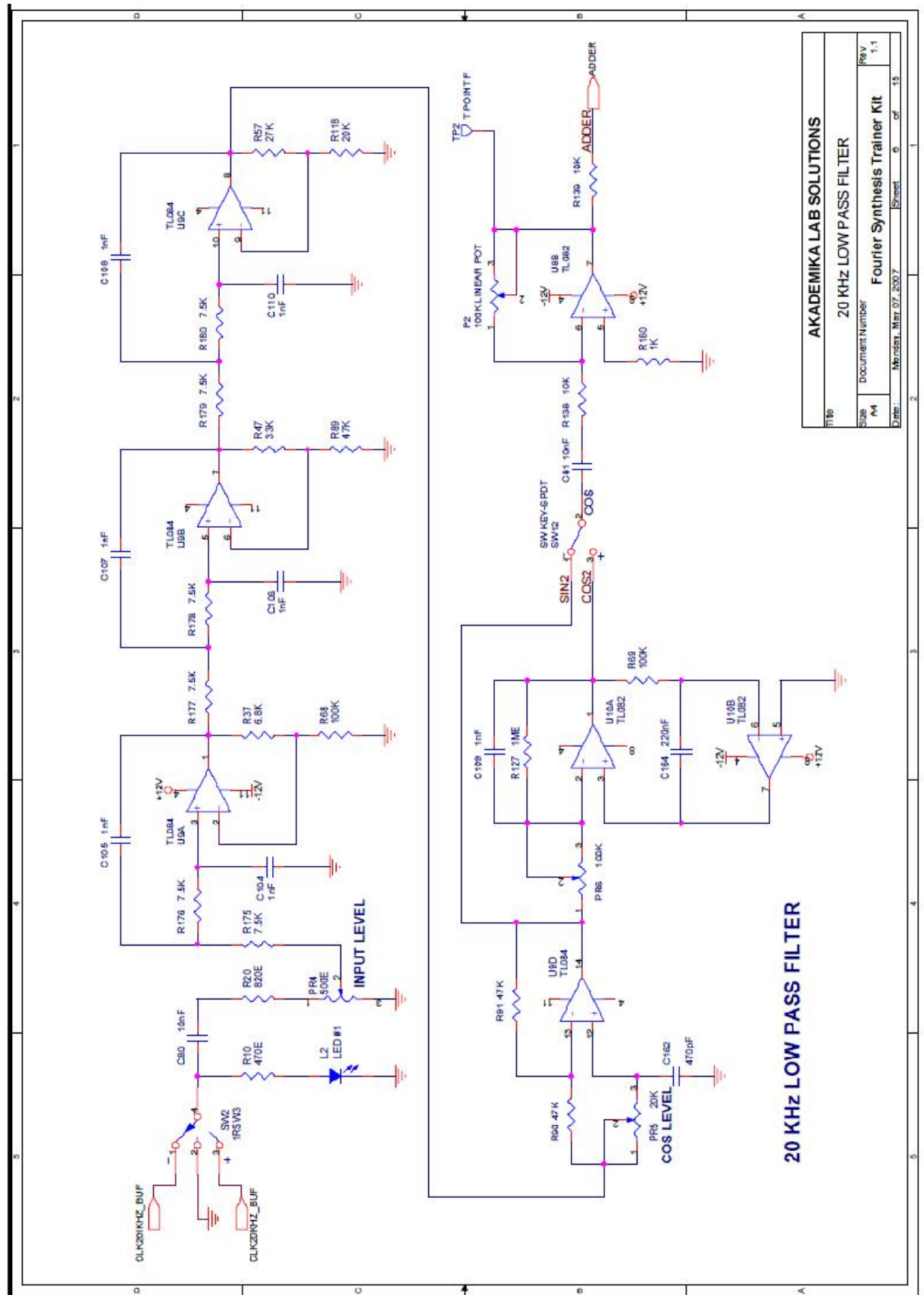


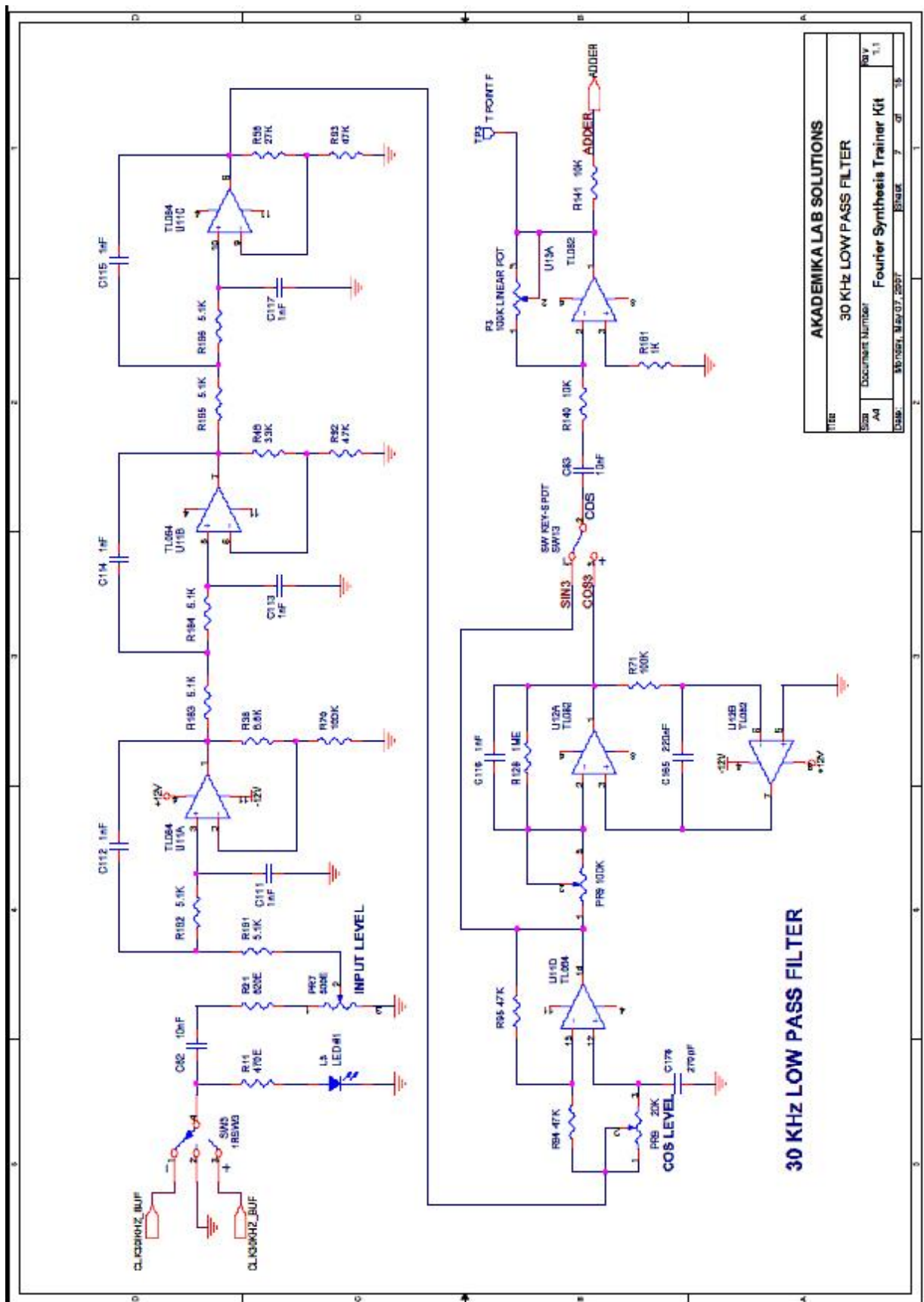
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70 KHz LOW PASS FILTER			
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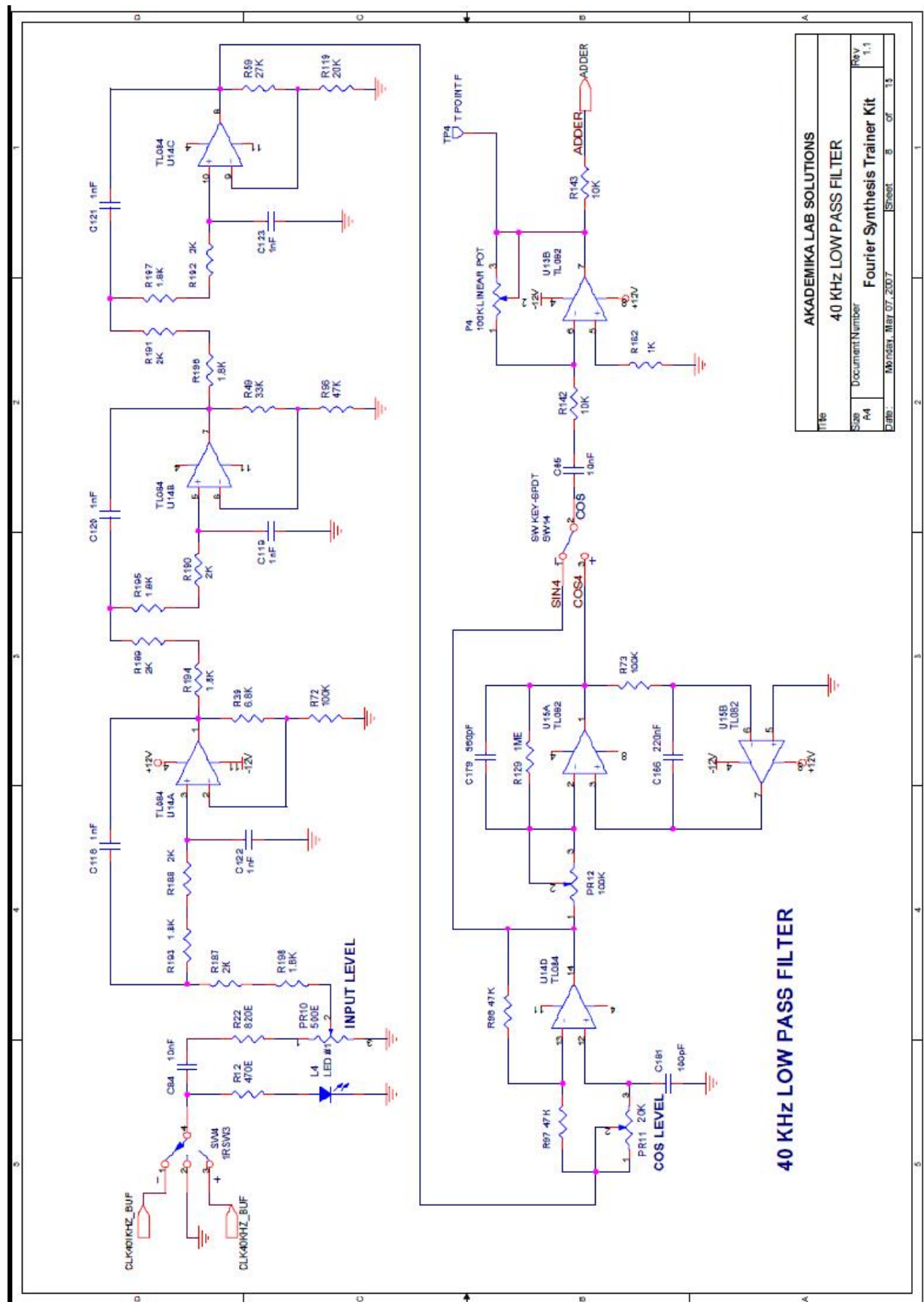


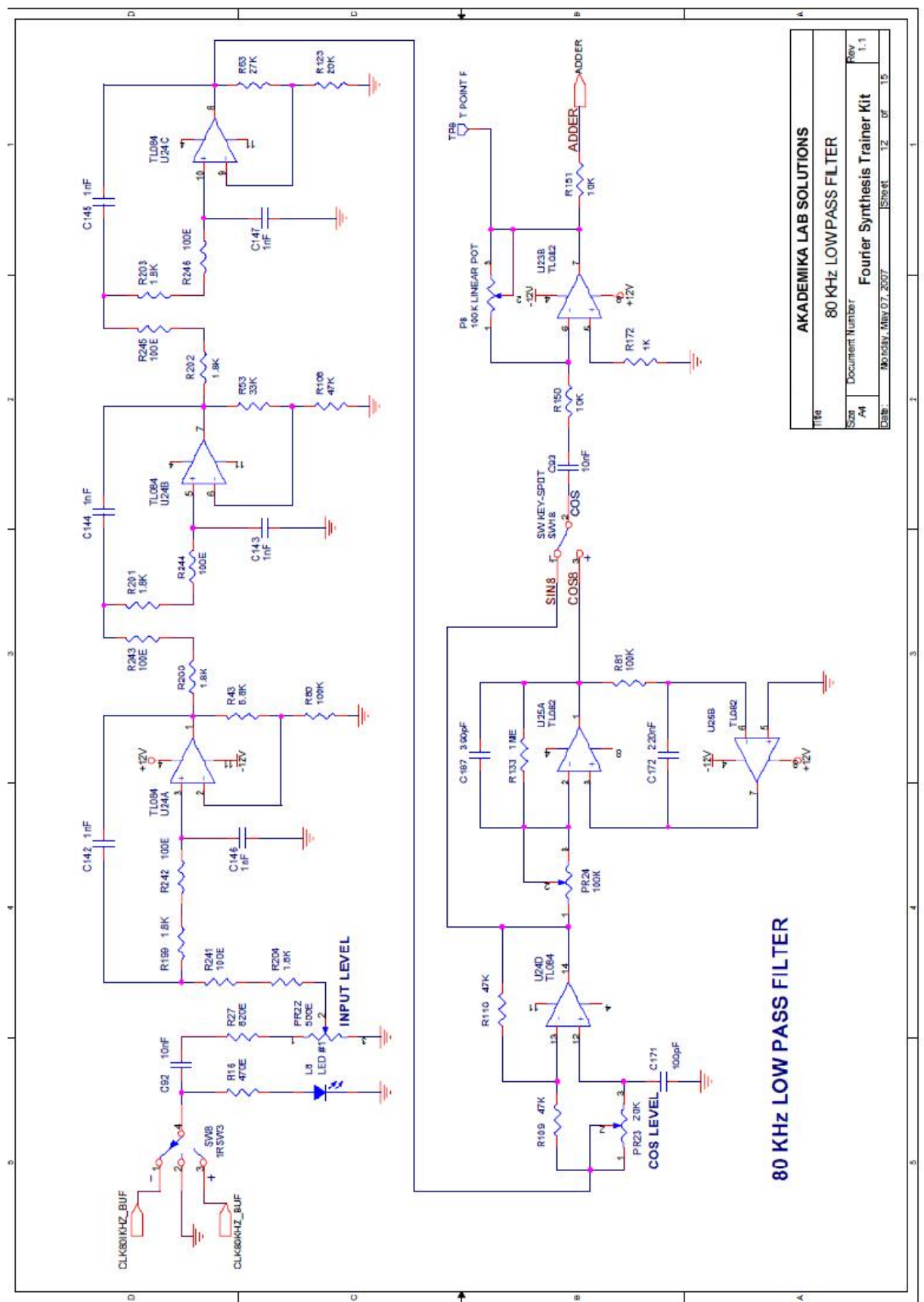


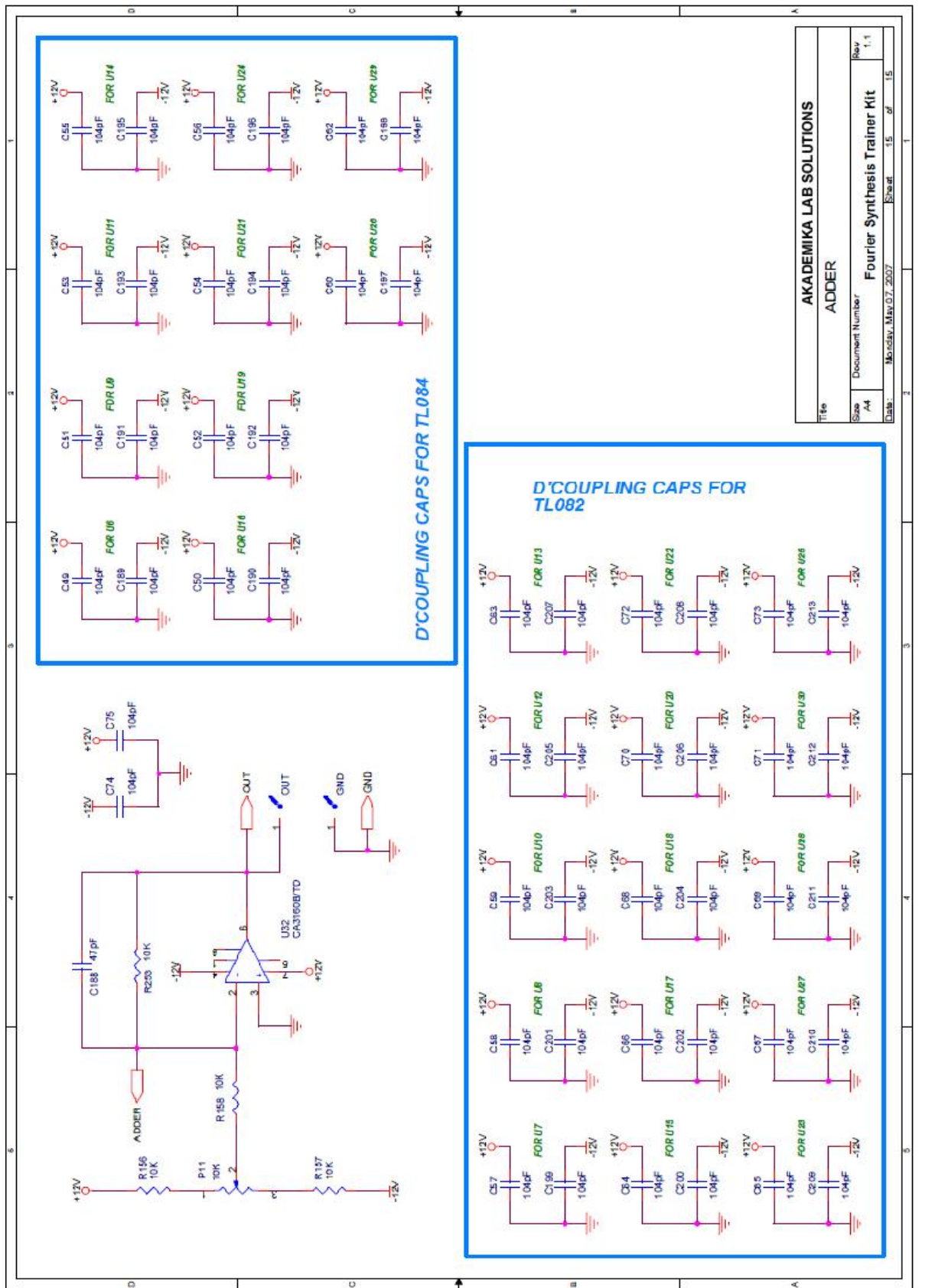


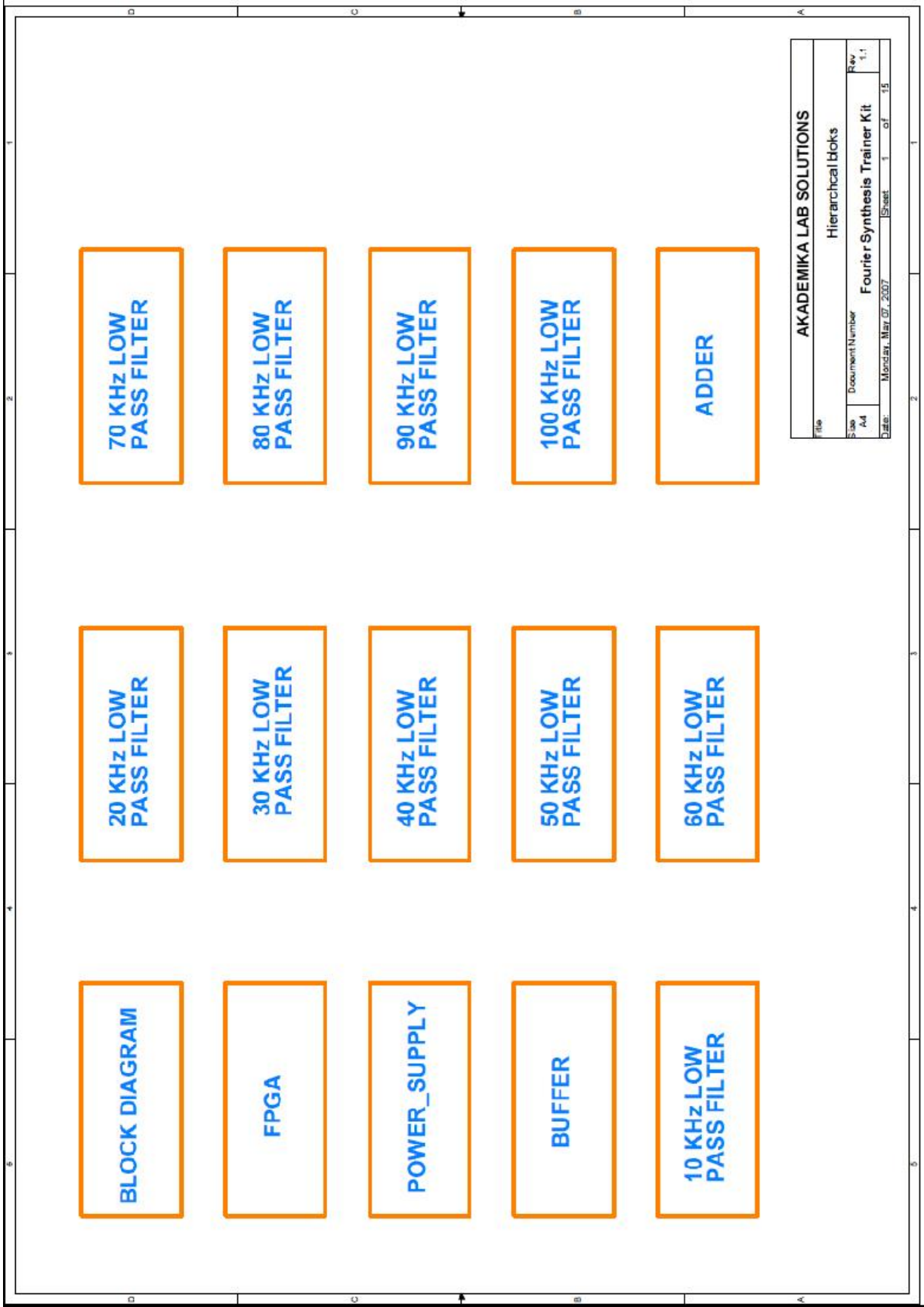


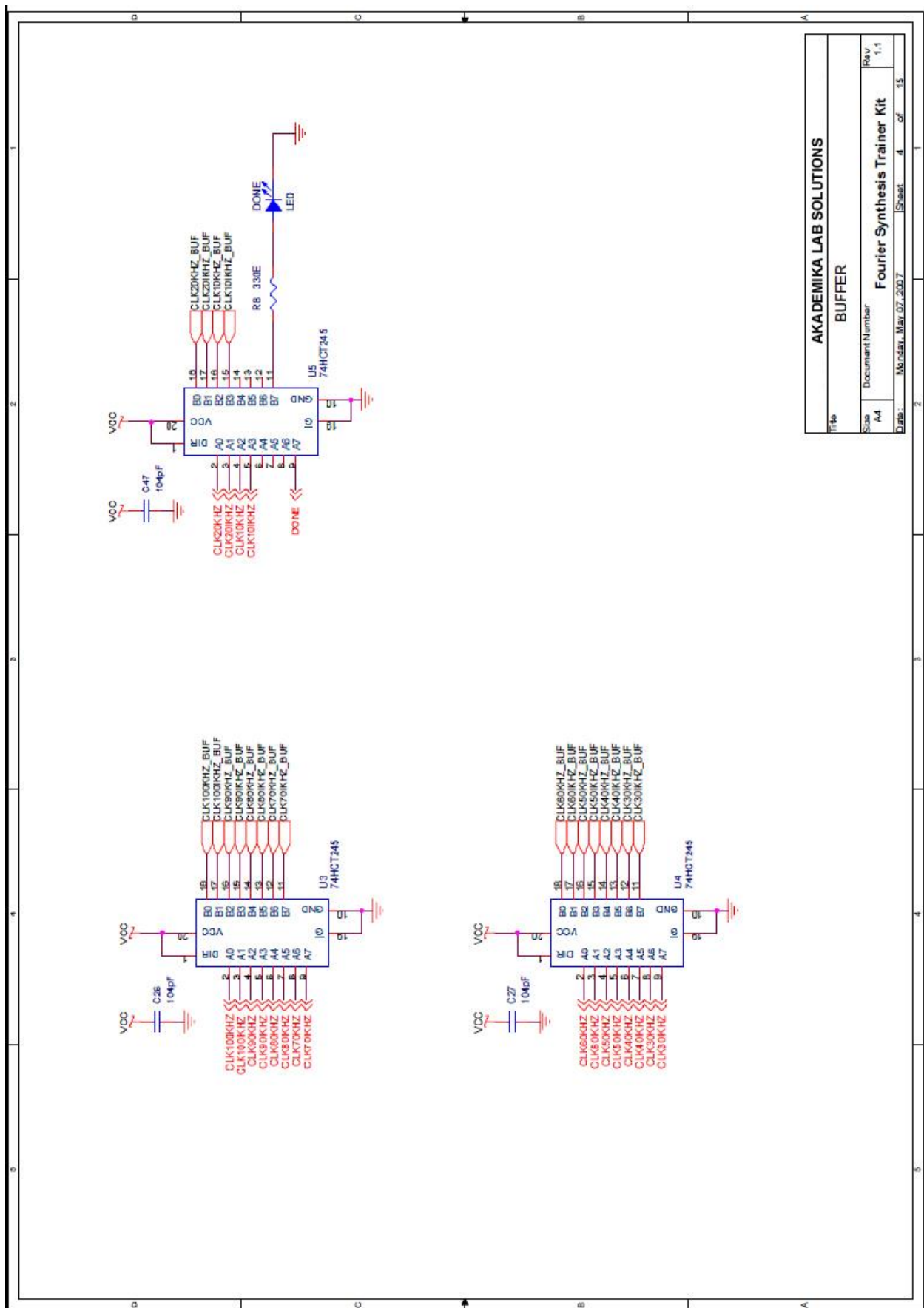




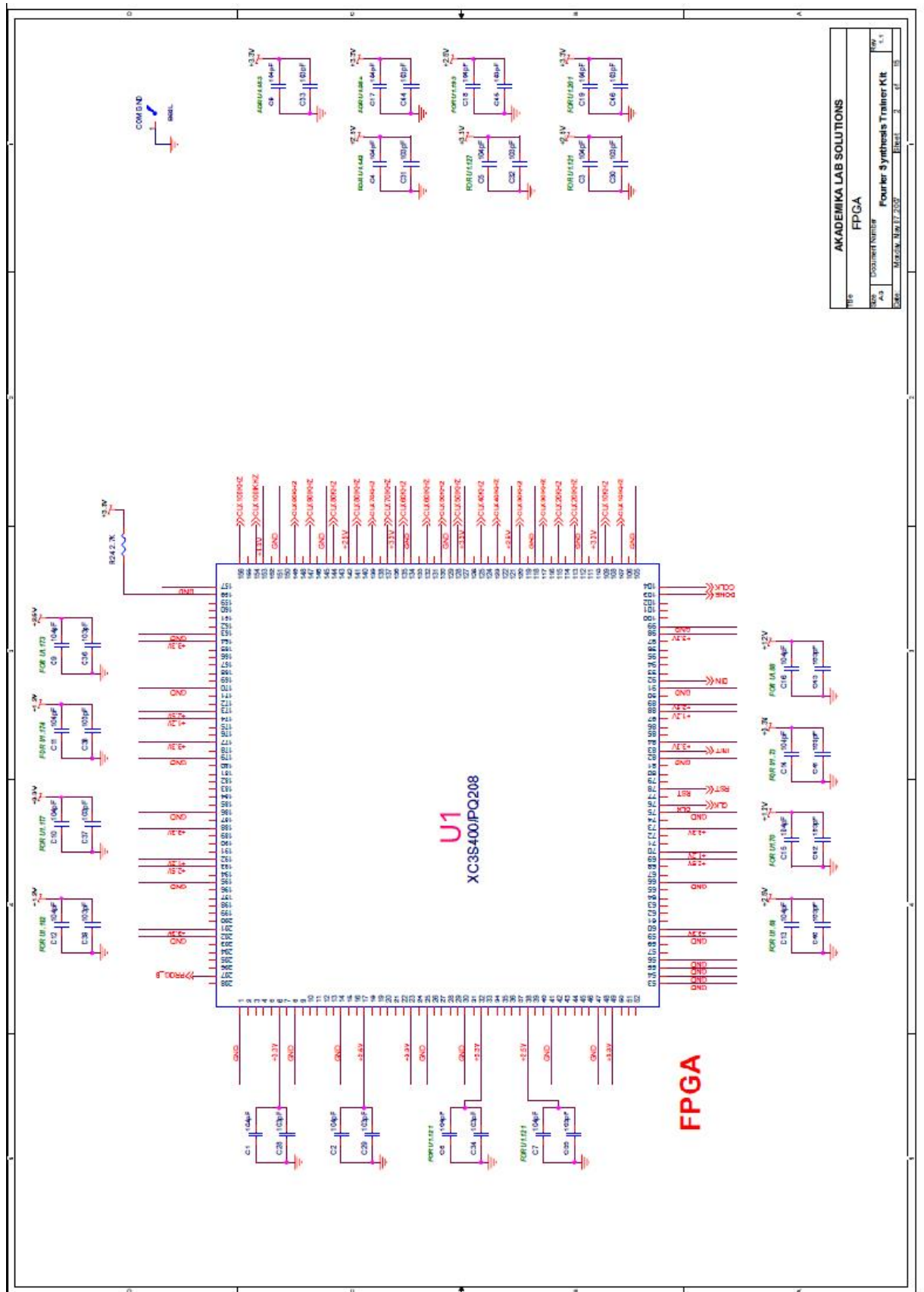


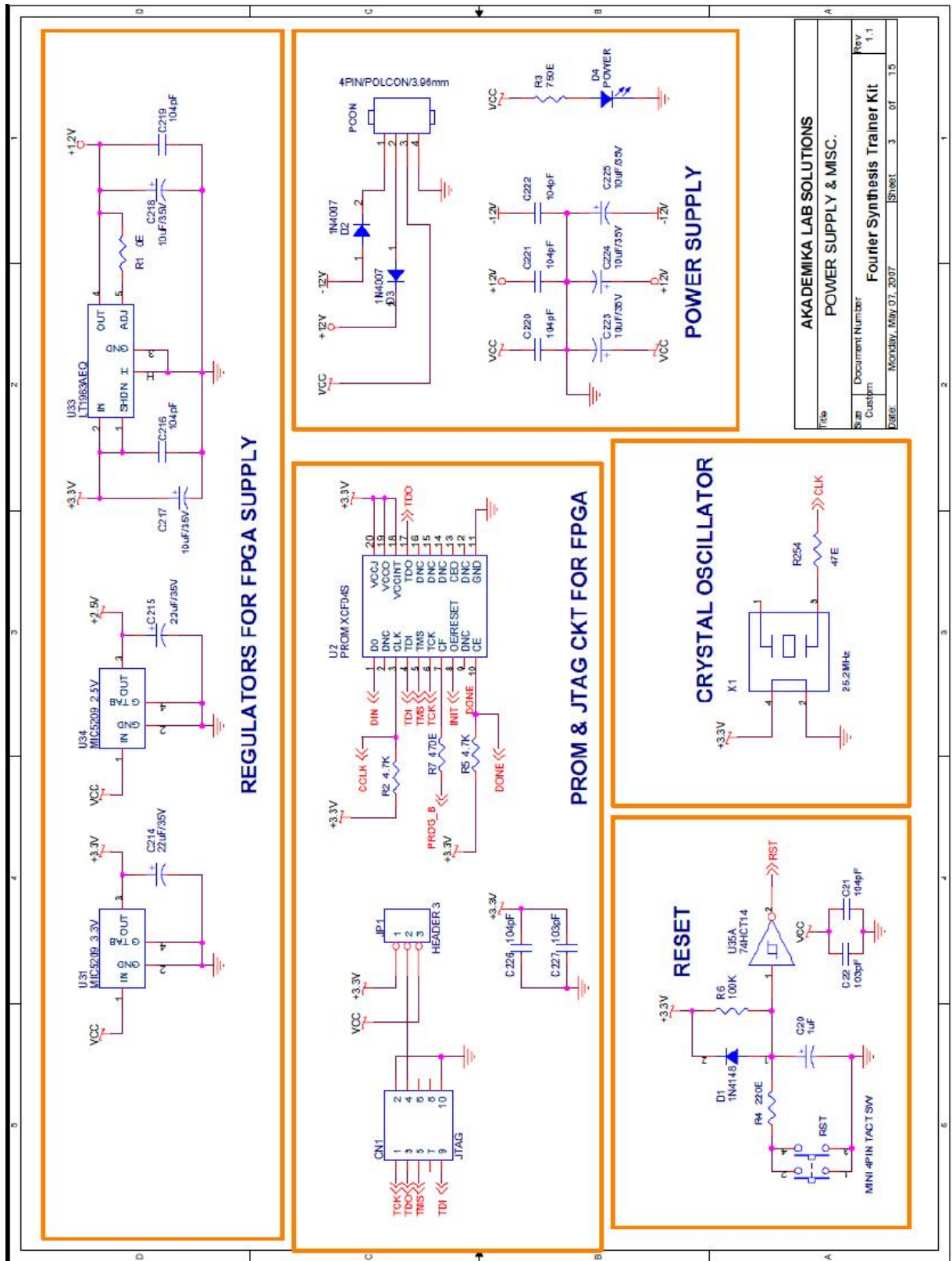






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EXPERIMENT NO.1

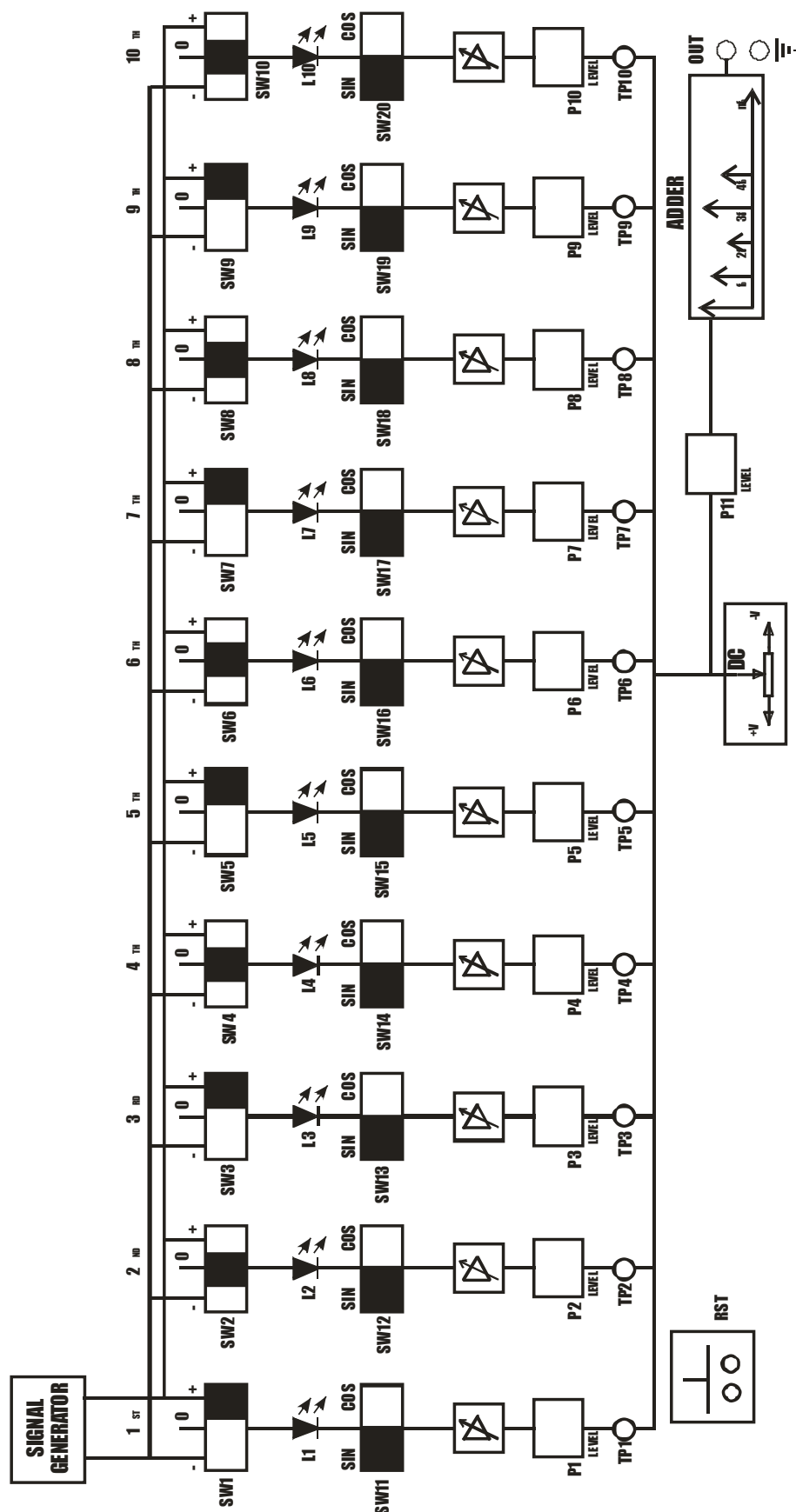


FIG. 1.1

SQUARE WAVE SYNTHESIS

EXPERIMENT NO.1

NAME:-

To Study of Square Waveform Synthesis

OBJECTIVE:-

The objective of this experiment is to study square wave synthesis

THEORY:-

Square Wave Synthesis

A square-wave spectrum is made of the sum of all the harmonics being odd of the fundamental, with decreasing amplitude according to a said. In other words, the square wave can be obtained by summing up the infinite sine waves as per the following relation:

$$S(t) = \frac{\sin(2\pi F.t)}{1} + \frac{\sin(2\pi.3.F.t)}{3} + \frac{\sin(2\pi.5.F.t)}{5} + \dots$$

EQUIPMENTS:-

- Fourier Synthesis Kit
- Patch chord
- Experimental manual
- Power supply
- Oscilloscope

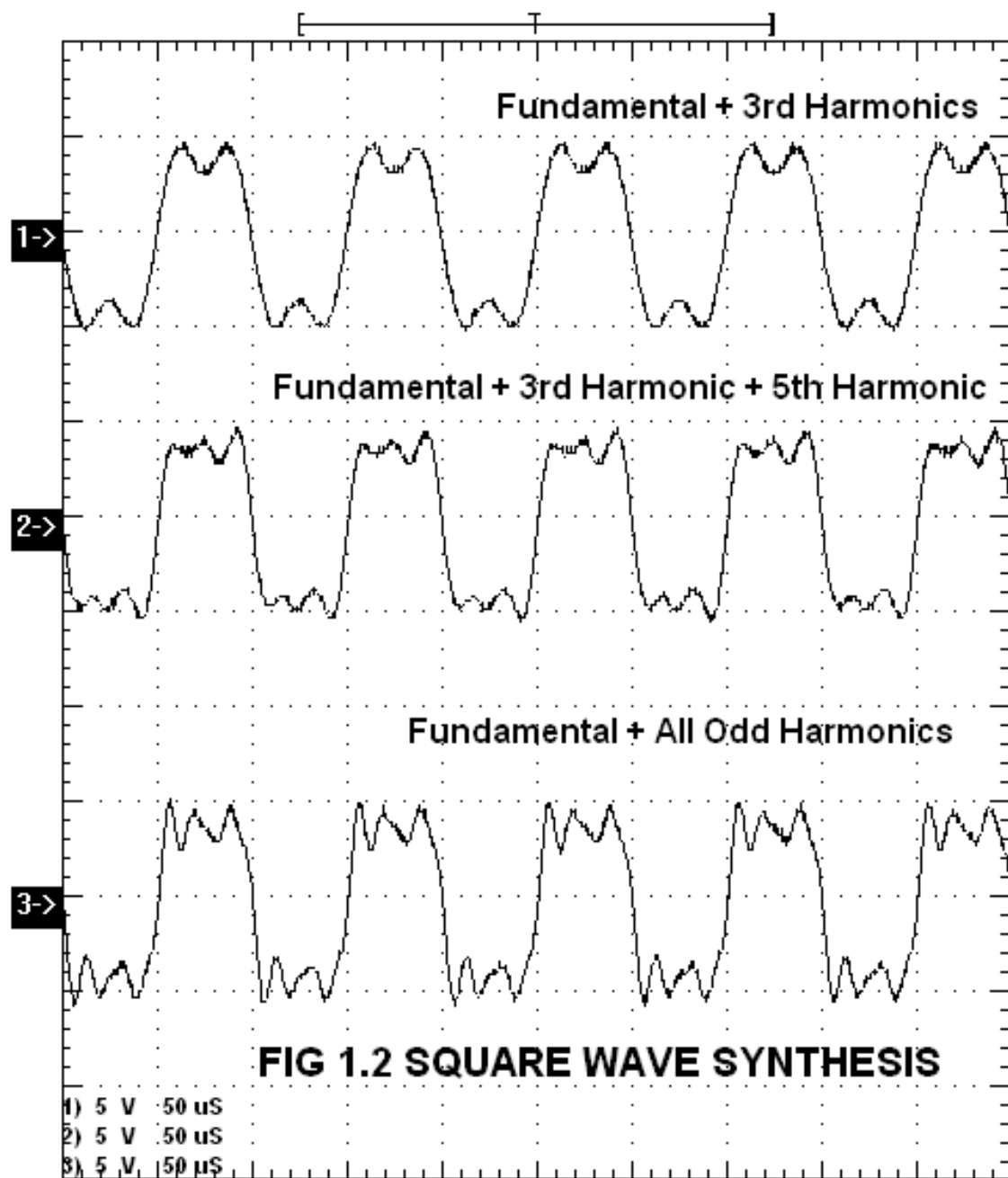
PROCEDURE:-

- Connect power supply with proper polarity to the kit & Switch on the power supply.
- by referring fig.1.1 set Odd harmonics (1/3/5/7/9): three-way switches “-/0/+” on “+” and three-way switches “sin/cos” on “sin”
- Even harmonics (2/4/6/8) : two-way switches “-/0/+” on “0”
- Connect the oscilloscope with the amplifier output of the fundamental (1st) and adjust the amplitude at 10Vpp

- Connect the oscilloscope with the output of the 3rd – harmonic amplifier (3rd) and adjust the amplitude at $10/3 \sim 3.3\text{Vpp}$.
- Connect the oscilloscope with the output of the 5th – harmonic amplifier (5th) and adjust the amplitude at $10/5 \sim 2\text{Vpp}$.
- Connect the oscilloscope with the output of the 7th – harmonic amplifier (7th) and adjust the amplitude at $10/7 \sim 1.4\text{Vpp}$.
- Connect the oscilloscope with the output of the 9th – harmonic amplifier (9th) and adjust the amplitude at $10/9 \sim 1.1\text{Vpp}$.
- Connect the oscilloscope at the OUT post and check that there is the signal corresponding to the component sum.
- Remove some harmonics (put the relating two-way switch on “0”) and check that by adding some harmonics the square wave is better approximated.

OBSERVATIONS:-

- Fundamental wave of 10 kHz at TP1
- 3rd harmonic at TP3
- 5th harmonic at TP5
- 7th harmonic at TP7
- 9th harmonic at TP9.
- Adder output at OUT post as shown in fig.1.2

WAVEFORM:-

EXPERIMENT NO.2

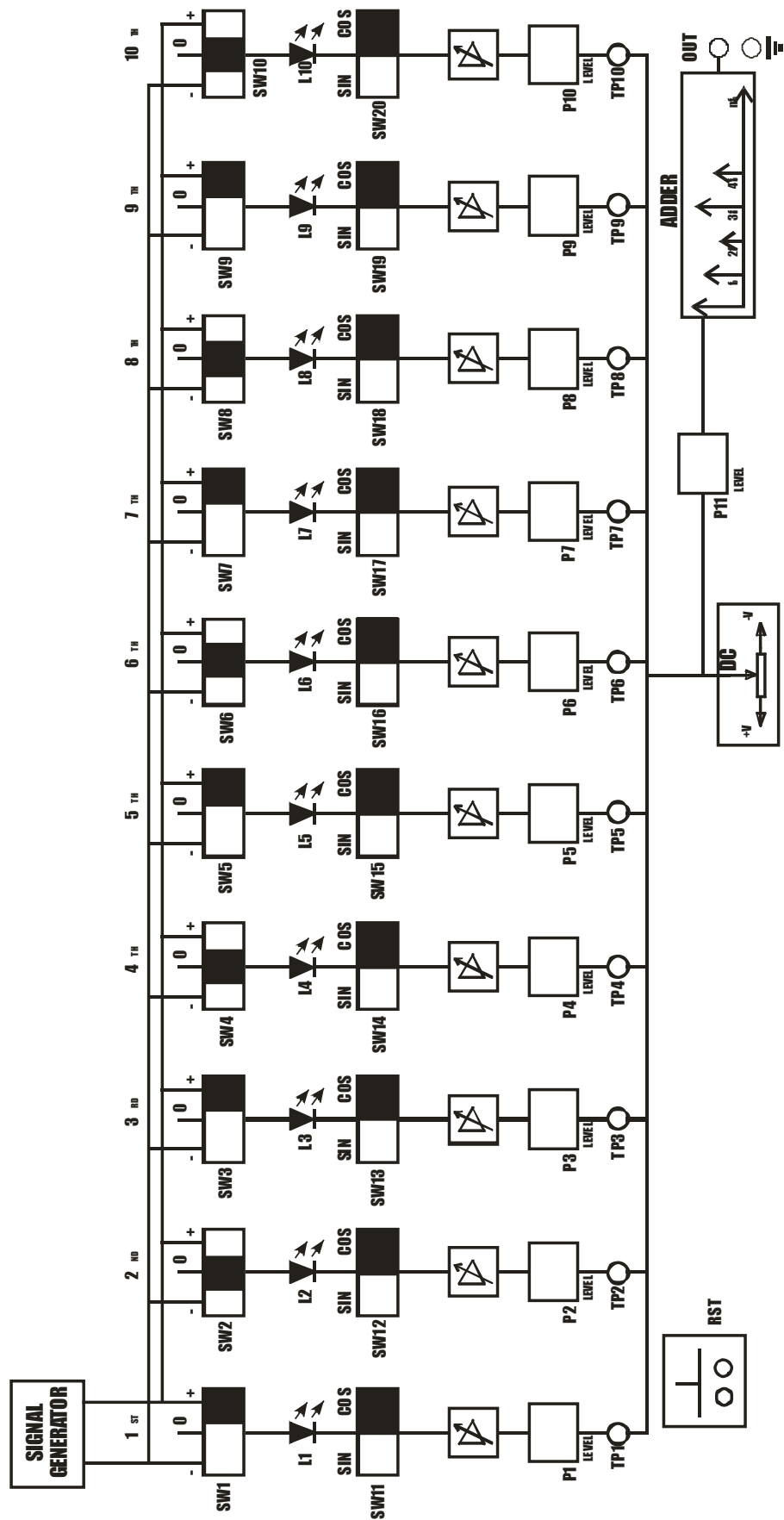


FIG. 2.1

EXPERIMENT NO.2

NAME:-

To Study of Triangular Waveform Synthesis

OBJECTIVE:-

The objective of this experiment is to study triangular wave synthesis

THEORY:-

Triangular Wave Synthesis

A triangular-wave spectrum is made of the sum of all the harmonics being odd of the fundamental, with decreasing amplitude according to a said law. In other words, the triangular wave can be obtained by summing up the infinite cosine waves as per the following relation:

The Fourier series development of a triangular wave gives:

$$S(t) = \frac{\cos(2\pi F.t)}{1^2} + \frac{\cos(2\pi.3.F.t)}{3^2} + \frac{\cos(2\pi.5.F.t)}{5^2} + \dots$$

EQUIPMENTS:-

- Fourier Synthesis Kit
- Patch chord
- Experimental manual
- Power supply
- Oscilloscope

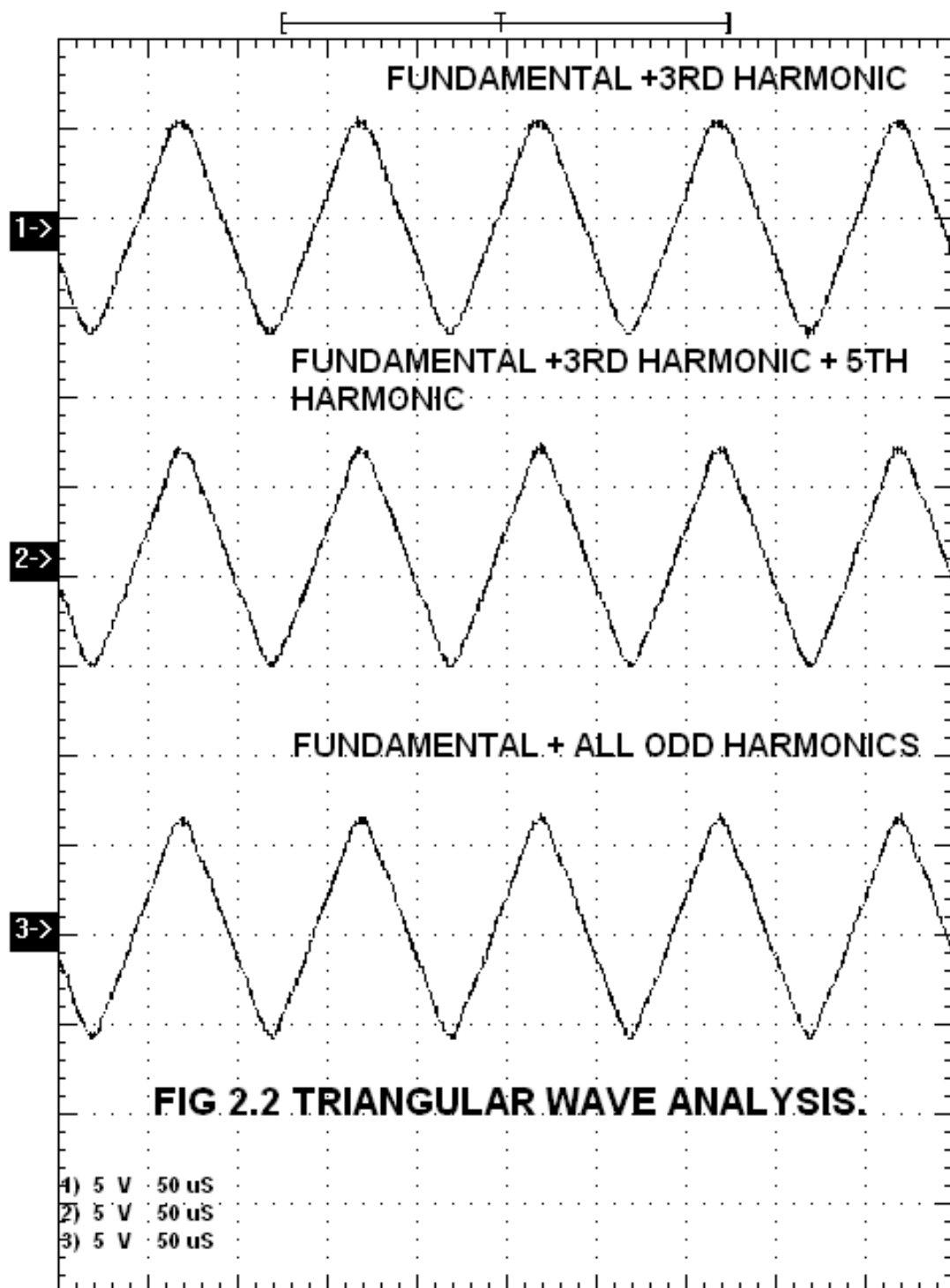
PROCEDURE:-

- Connect power supply with proper polarity to the kit & Switch on the power supply.
- by referring fig 2.1, set Odd harmonics (1/3/5/7/9) : three-way switches “-/0/+” on “+” and two-way switches “sin/cos” on “cos”
- Even harmonics (2/4/6/8) : two-way switches “-/0/+” on “0”
- Adjust the fundamental amplitude (1st) at 10Vpp.

- Adjust the amplitude of the odd harmonics as follows:
 $3^{\text{rd}} \ 10/3^2 \approx 1.1 \text{ Vpp}$
 $5^{\text{th}} \ 10/5^2 \approx 0.4 \text{ Vpp}$
 $7^{\text{th}} \ 10/7^2 \approx 0.2 \text{ Vpp}$
 $9^{\text{th}} \ 10/9^2 \approx 0.1 \text{ Vpp}$
- Connect oscilloscope at OUT post and check that there is the signal corresponding to the sum of the components.
- Remove some harmonics (put the relating two-way switches on "0") and check that when adding some harmonics the triangular wave is better approximated.

OBSERVATIONS:-

- Fundamental wave of 10 kHz at TP1
- 3^{rd} harmonic at TP3
- 5^{th} harmonic at TP5
- 7^{th} harmonic at TP7
- 9^{th} harmonic at TP9.
- Adder output at OUT post as shown in fig.2.2

WAVEFORM:-

EXPERIMENT NO.3

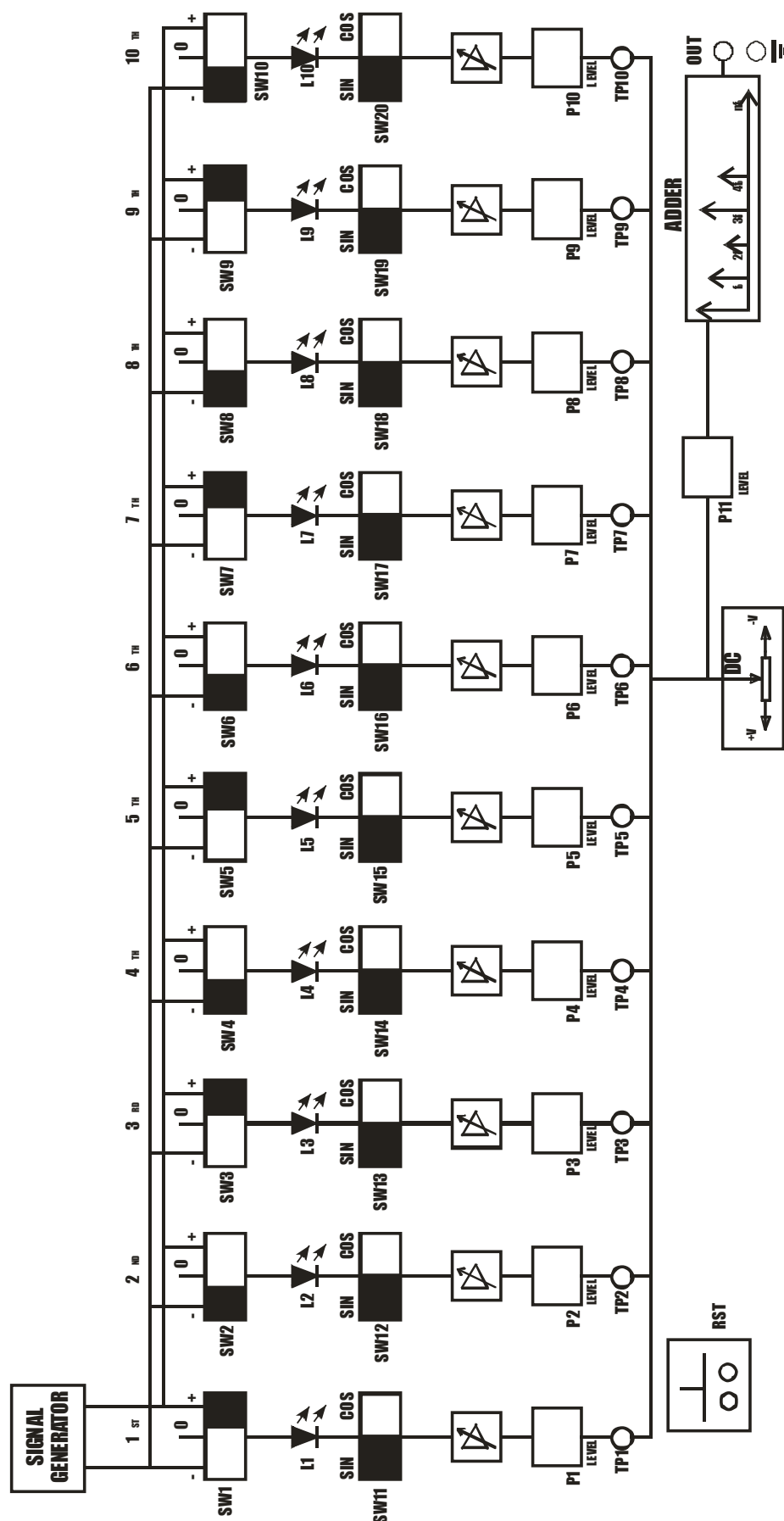


FIG. 3.1

SAW-TOOTH WAVE SYNTHESIS

EXPERIMENT NO.3

NAME:-

To Study of Saw-Tooth Waveform Synthesis

OBJECTIVE:-

The objective of this experiment is to study saw-tooth wave synthesis

THEORY:-

Saw-Tooth Wave Synthesis

A saw-tooth spectrum is made of the sum of all the harmonics being odd and even of the fundamental, with decreasing amplitude according to a said law. In other words, the saw-tooth wave can be obtained by summing up the infinite sine waves as per the following relation:

The Fourier-series development of saw-tooth wave gives:

$$S(t) = \frac{\sin(2\pi F.t)}{1} - \frac{\sin(2\pi.2.F.t)}{2} + \frac{\sin(2\pi.3.F.t)}{3} - \frac{\sin(2\pi.4.F.t)}{4} + \frac{\sin(2\pi.5.F.t)}{5} - \frac{\sin(2\pi.6.F.t)}{6} \dots$$

EQUIPMENTS:-

- Fourier Synthesis Kit
- Patch chord
- Experimental manual
- Power supply
- Oscilloscope

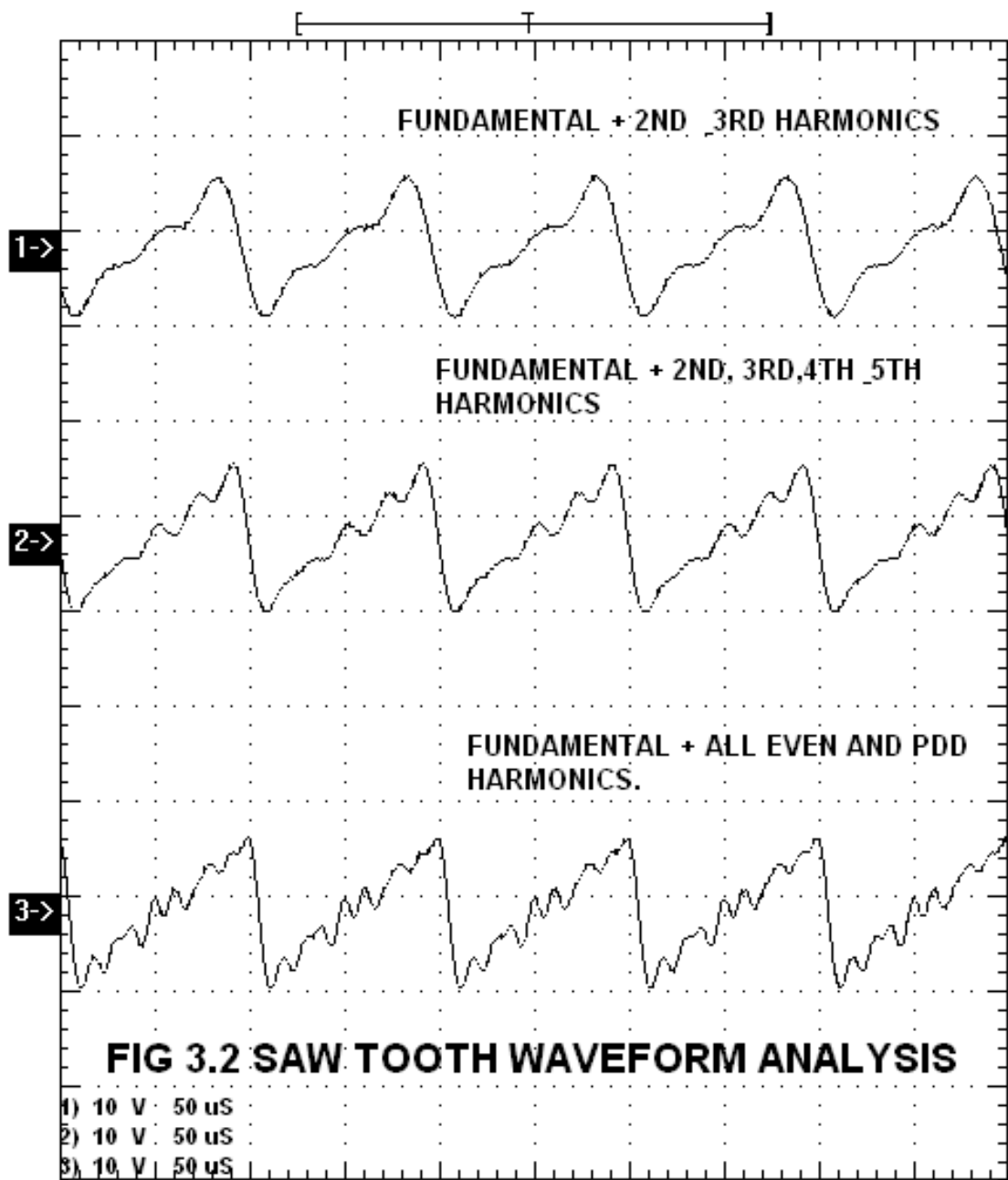
PROCEDURE:-

- Connect power supply with proper polarity to the kit & Switch on the power supply.
- by referring fig 3.1, set Odd harmonics (1/3/5/7/9): three-way switches “-/0/+” on “+” and two-way switches “sin/cos” on “sin”

- Even harmonics (2/4/6/8): two-way switches “-/0/+” on “-” and two-way switches “sin/cos” on “sin”
- Adjust the fundamental amplitude (1st) at 10Vpp.
- Adjust the harmonic amplitude as follows:
$$2^{\text{nd}} \ 10/2 = 5 \ \text{Vpp}$$
$$3^{\text{rd}} \ 10/3 \approx 3.3 \ \text{Vpp}$$
$$4^{\text{th}} \ 10/4 \approx 2.5 \ \text{Vpp}$$
$$5^{\text{th}} \ 10/5 = 2 \ \text{Vpp}$$
- Connect the oscilloscope at OUT post, and check that there is that signal corresponding to the component sum
- Remove some harmonics (put the relating two-way switch on “0”) and check that by adding some harmonics the triangular wave is better approximated.

OBSERVATIONS:-

- Fundamental wave of 10 kHz at TP1
- 2nd harmonic at TP2
- 3rd harmonic at TP3
- 4th harmonic at TP4
- 5th harmonic at TP5
- 6th harmonic at TP6
- 7th harmonic at TP7
- 8th harmonic at TP8
- 9th harmonic at TP9
- 10th harmonic at TP10.
- Adder output at OUT post as shown in fig.3.2

WAVEFORM:-

EXPERIMENT NO.4

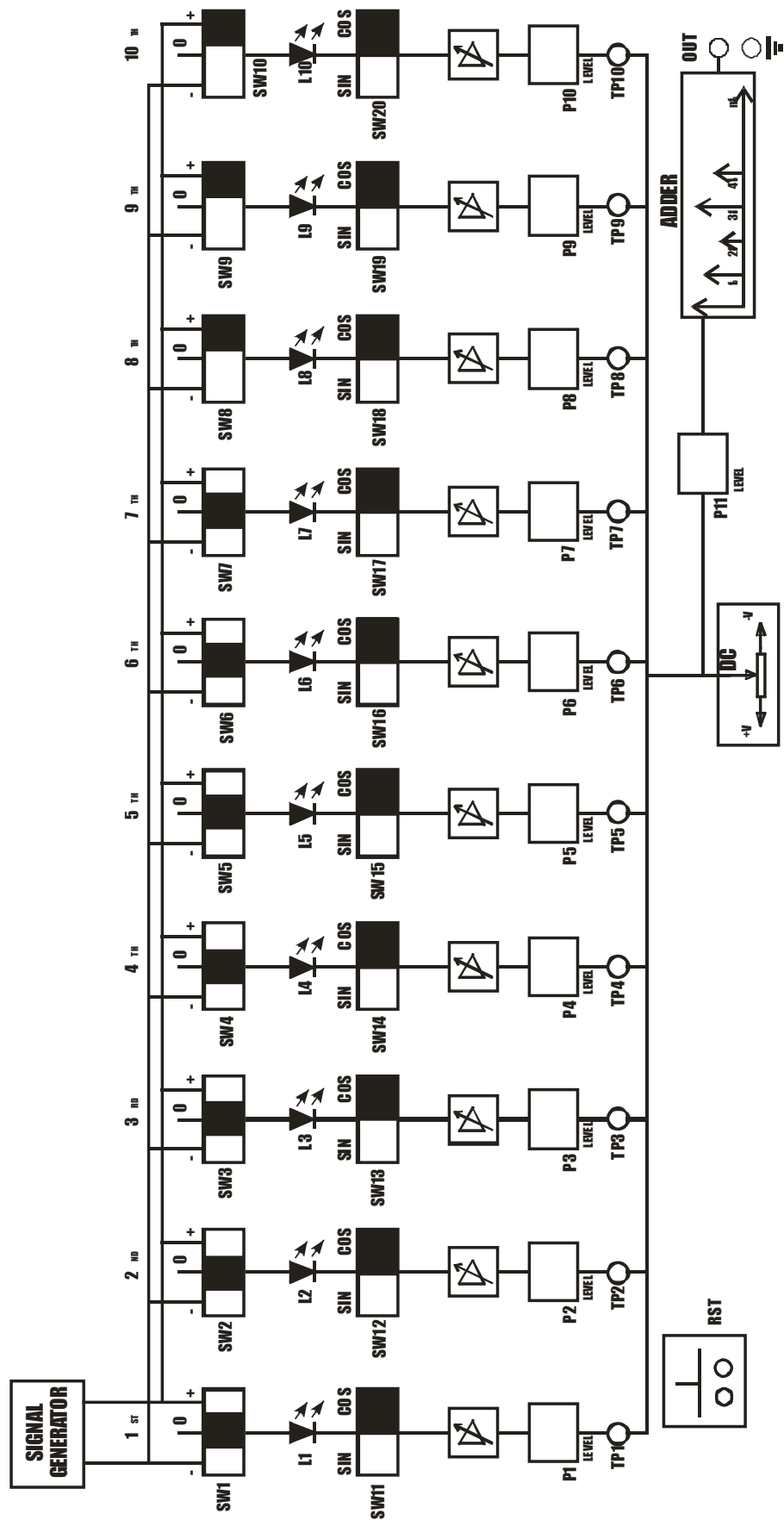


FIG. 4.1

AM MODULATED WAVE SYNTHESIS

EXPERIMENT NO.4

NAME:-

To Study Of Amplitude Modulated Signal Synthesis

OBJECTIVE:-

The objective of this experiment is to study amplitude modulated signal synthesis

THEORY:-

Amplitude- Modulated Signal Synthesis

Be a sine wave modulating at frequency f:

$$V_m(t) = \cos(2\pi f.t)$$

And a sine signal carrying at a higher frequency

$$V_C(t) = m \cdot \cos(2\pi F.t)$$

The amplitude-modulated signal shows the following shape:

$$V_M(t) = [1 + m \cdot \cos(2\pi f.t)] \cdot \cos(2\pi F.t)$$

By using simple trigonometric operations the former relation becomes:

$$V_M(t) = \cos(2\pi F.t) + \frac{m}{2} \cdot \cos[2\pi(F - f)t] + \frac{m}{2} \cdot \cos[2\pi(F + f)t]$$

Which we deduct from that the amplitude-modulated signal consists of three sine components:

$$\cos(2\pi F.t)$$

Carrier

$$\frac{m}{2} \cdot \cos[2\pi(F - f)t]$$

Lower sideband

$$\frac{m}{2} \cdot \cos[2\pi(F + f)t]$$

Upper sideband

If: $F = 9.F$ and $f = F_0$

The relation expressing the amplitude-modulated signal becomes:

$$V_M(t) = \cos(2\pi \cdot 9F_0 \cdot t) + \frac{m}{2} \cdot \cos[2\pi \cdot 8F_0 \cdot t] + \frac{m}{2} \cdot \cos[2\pi \cdot 10F_0 \cdot t]$$

Using frequency $9F_0$ as a carrier and frequencies $8F_0$ and $10F_0$ as sidebands, the waveform corresponding to amplitude modulation can be synthesized

EQUIPMENTS:-

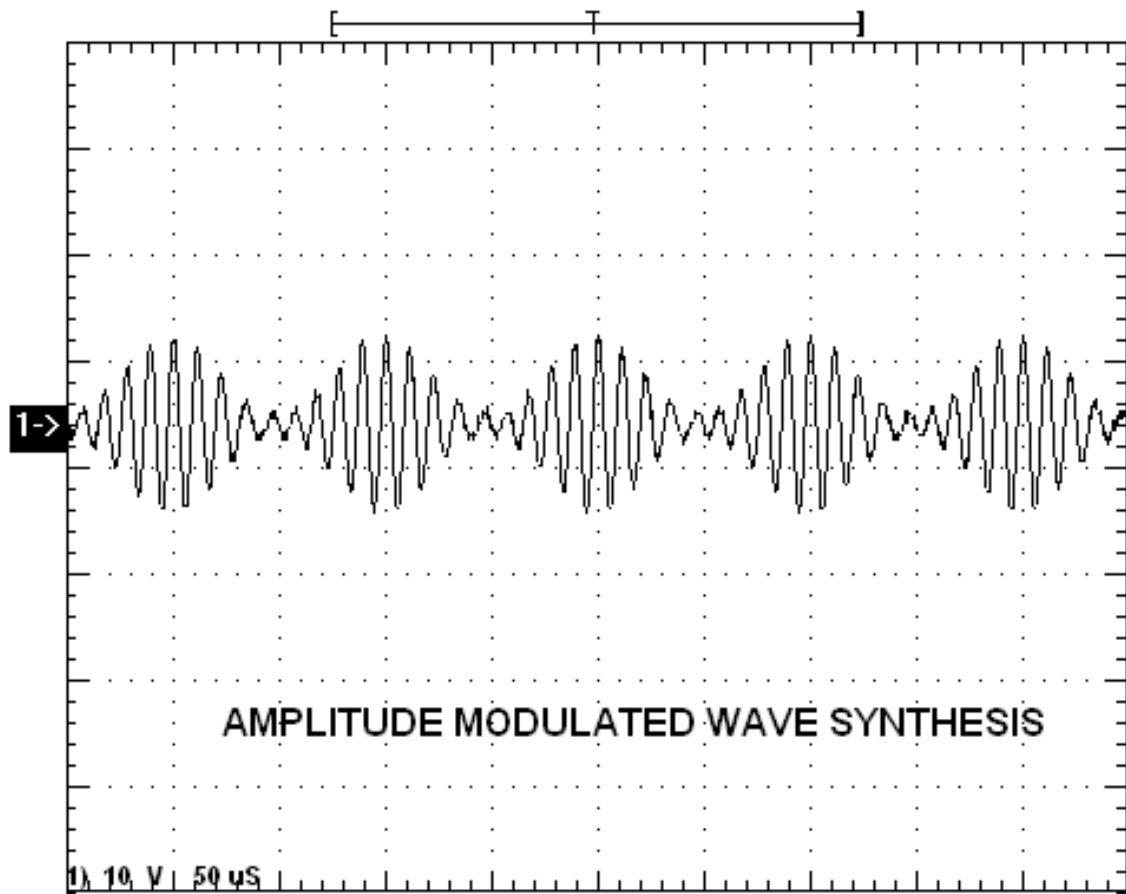
- Fourier Synthesis Kit
- Patch chord
- Experimental manual
- Power supply
- Oscilloscope

PROCEDURE:-

- Connect power supply with proper polarity to the kit & Switch on the power supply.
- by referring fig 4.1, set harmonics (8^{th} , 9^{th} and 10^{th}): three-way switches “-/0/+” on “+” and two-way switches “sin/cos” on “cos”
- Adjust the fundamental amplitude (8^{th}) at 5Vpp.
- Adjust the fundamental amplitude (9^{th}) at 10Vpp
- Adjust the fundamental amplitude (10^{th}) at 5Vpp
- Connect the oscilloscope at OUT post, and check that there is the signal corresponding to the component sum

OBSERVATIONS:-

- 8^{th} harmonic at TP8
- 9^{th} harmonic at TP9
- 10^{th} harmonic at TP10
- Adder output at OUT post as shown in fig.4.2

WAVEFORM:-**FIG 4.2**